

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a deep blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

LECTURE 7.2

Fundamental PDEs

Five equations, one loop. Each is defined once, with the conditions that make it well posed, and each introduces exactly one new difficulty for a physics-informed network.

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Deep Learning for Engineering · 2026

Poisson · heat · wave · Helmholtz · Burgers

FIVE EQUATIONS, ONE DIFFICULTY EACH

Poisson

elliptic — the reference

heat

parabolic — time as a coordinate

wave

hyperbolic — two initial conditions

Helmholtz

eigenvalue — a trivial solution

Burgers

nonlinear — a shock

The loop of the last lecture does not change. What changes, row by row, is the operator and what it needs to be well posed.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Classification** — the discriminant; elliptic, parabolic, hyperbolic, and what each needs
- 2 **Poisson** — the reference problem; mixed conditions; a learned boundary factor
- 3 **Heat** — first order in time; the space–time block; a hard initial condition
- 4 **Wave** — second order in time; two initial conditions; periodic activations
- 5 **Helmholtz** — an eigenvalue unknown too; the trivial solution; orthogonality
- 6 **Burgers** — nonlinearity; the shock; discovering the equation from data
- 7 **Verification** — non-dimensionalise first; four checks a PINN cannot do for you
- 8 **Challenges** — a symptom and a fix for each equation

ASSUMED, FROM THE LAST LECTURE

the loop	network, autograd, residual, step
the terms	PDE, DBC, NBC, IC, IV, and weights
enforcement	soft as a term, hard as a factor
training	Adam then L-BFGS; refine; normalise

Nothing in that loop changes today. Only the operator and its admissible conditions do, and the exercise is built on the same three modules as Ex_07.1.

Recap

THE TEN QUESTIONS OF L7.1 — PHYSICS-INFORMED NEURAL NETWORKS

- 1** What kind of network is a PINN, what function does it compute, and how does its loss differ from a supervised one? *Raissi et al. (2019); Liu ch. 2.1*
- 2** What does the universal approximation theorem promise for the ansatz, and what does it not? *Cybenko (1989); Liu ch. 2.1*
- 3** Why must the activation be smooth, stated from the order of the equation, and what does ReLU do to a Laplacian? *Liu ch. 4.7*
- 4** What is P^* , how does it differ from the weight count, and what does it say about the sample size? *Liu ch. 2.5.2, eq. (2.19)*
- 5** How does autograd produce $\partial^2 u / \partial x^2$, why is `create_graph` needed, and what does a finite difference cost instead? *Liu ch. 2.2.2; Ex_07 notebook 00*
- 6** Which conditions make Poisson, heat and wave well posed, and which operator does each add to the loss? *Liu ch. 2.2, eq. (2.2) – (2.5)*
- 7** Soft against hard enforcement of a Dirichlet condition: what does each cost, and what did the measurement show? *Liu ch. 2.5.3; Ex_07 notebooks 01 – 02*
- 8** What is the weight between two loss terms, physically, and what does gradient-norm balancing do? *Liu ch. 2.2.3; Wang et al. (2021)*
- 9** Collocation, stencil or energy: what does each ask of the network, and what does each cost? *Liu ch. 2.6 – 2.8*
- 10** Why Adam then L-BFGS, when do you switch, and what is spectral bias? *Liu ch. 2.7.5; Xu et al. (2019)*

PDE Classification

SECOND ORDER, LINEAR, ONE DISCRIMINANT

$$A u_{xx} + B u_{xy} + C u_{yy} + D u_x + E u_y + F u = G$$

$$\Delta = B^2 - 4AC$$

the sign of Δ sorts every second-order linear equation into one of three types, and the type says which conditions make the problem well posed

	Δ	example	character	conditions
elliptic	< 0	Poisson	equilibrium, smooth	boundary only
parabolic	$= 0$	heat	diffusive, smoothing	boundary, one initial
hyperbolic	> 0	wave	finite speed, preserving	boundary, two initial

- **elliptic** — no time; every point feels every boundary at once
- **parabolic** — first order in time; disturbances spread and smooth
- **hyperbolic** — second order in time; information travels at speed c

THE FIVE EQUATIONS OF TODAY

	type	what is new
Poisson	elliptic	the reference
heat	parabolic	time as a coordinate
wave	hyperbolic	two initial conditions
Helmholtz	eigenvalue	an unknown scalar too
Burgers	nonlinear	a shock

Helmholtz and Burgers are outside the three types: an eigenvalue problem, and a nonlinear one. They are here because the application blocks need them, and each brings a failure the first three do not.

Poisson Equation

KIND, FUNCTION, CONDITIONS, APPLICATIONS

- **kind** — elliptic; no time, every point feels every boundary
- **function** — a steady balance with a distributed source
- **conditions** — a value or a flux at every boundary point, no gaps
- **applications** — steady conduction, electrostatics, potential flow

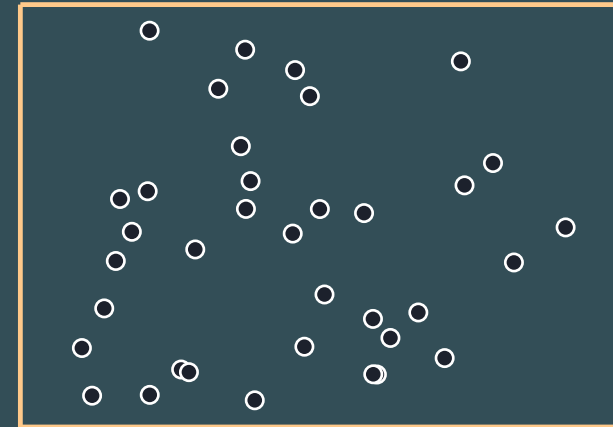
$$u_{xx} + u_{yy} + f(x, y) = 0$$

second order, two dimensions, no time: *tanh* and two autograd passes suffice

$$u = \bar{u} \text{ on } \partial\Omega_D, \quad \frac{\partial u}{\partial n} = \bar{t} \text{ on } \partial\Omega_N$$

Dirichlet on one part of the boundary, Neumann on the rest; together they cover all of it

A MIXED PROBLEM



solid: Dirichlet · dashed: Neumann

Dirichlet

fix the state

Neumann

fix the flux

Neumann on every wall: fixed only up to a constant.

Learned Boundary Factor

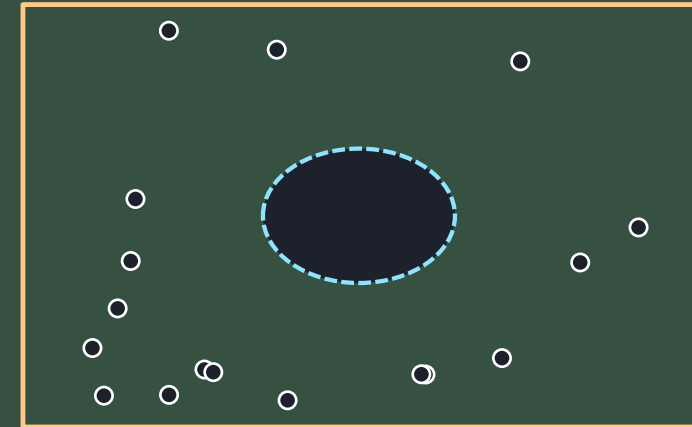
HARD ENFORCEMENT ON A DOMAIN WITH NO POLYNOMIAL

$$u_{\text{trial}}(x, y) = \mathcal{N}_{\hat{v}}^D(x, y) \mathcal{N}_{\hat{w}}(x, y)$$

a second small network, trained to vanish on the Dirichlet edge, then frozen

- a rectangle has a polynomial factor; a plate with a hole does not
- train the factor first: zero on the outer edge, positive inside
- freeze it, and use it exactly as the polynomial was
- the boundary term is gone again; the composite loss is one term
- the T3 mesh integrates; the network is still the solution

A PLATE WITH AN ELLIPTICAL HOLE



outer edge: Dirichlet · hole: Neumann

The factor must vanish on the outer edge and stay positive around the hole. No single polynomial does both, so it is learned; the Neumann hole stays a loss term.

Heat Equation

KIND, FUNCTION, CONDITIONS, APPLICATIONS

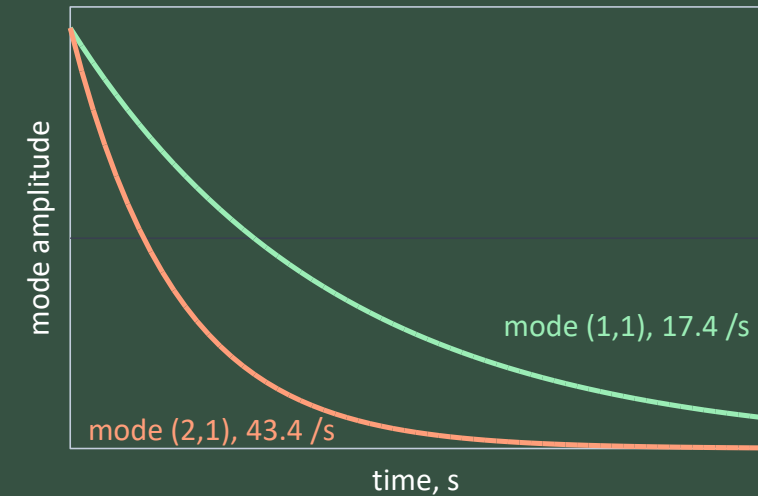
- **kind** — parabolic; first order in time
- **function** — diffusion: a field that smooths and spreads
- **conditions** — the boundary for all t , and one initial field
- **against Poisson** — one new term, a third input, time as an axis
- **applications** — transient conduction, a die after a power pulse

$$u_t = \alpha (u_{xx} + u_{yy}) + f(x, y, t)$$

α is the thermal diffusivity; set the time derivative to zero and Poisson returns

$$u(x, y, 0) = f_{IC}(x, y)$$

THE DIE'S TWO MODES DECAY



Ex_07.2's die: mode (m, n) decays at $(m^2 + n^2) \pi^2 \alpha / L^2$. The sharp feature dies two and a half times faster, so the field changes shape, not only size.

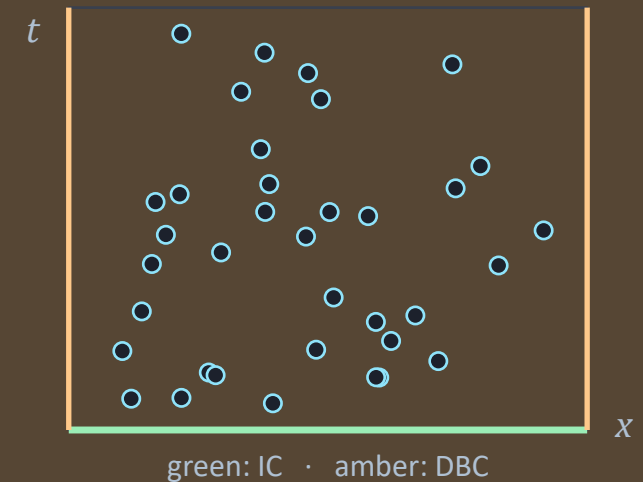
Space-Time Domain

A PINN DOES NOT MARCH THROUGH TIME

- t is one more input; the whole block is trained at once
- the solution is continuous in t , evaluable at any instant
- a departure from time stepping, where each step needs the last
- nothing requires early times to be right first
- the initial field is one slice, against a whole block of interior points

TALK · 2 MIN A time-marching solver cannot get $t = 0.3$ wrong before $t = 0.1$. A PINN can. What does that make possible, and what does it cost you?

$\Omega \times \text{TIME}$, SAMPLED AS ONE BLOCK



the slice one condition, few points

the block the residual, many points

Cyan is the residual. The initial slice is outvoted.

Hard Initial Condition

A TRIAL SOLUTION THAT ENFORCES THE IC AND THE WALLS

$$u_{\text{trial}} = (1 - t) f_{\text{IC}}(x, y) + t x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}(x, y, t)$$

at $t = 0$ the second term vanishes and the initial field is returned, for any weights

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{DBC}} + \mathcal{L}_{\text{IC}} \Rightarrow \mathcal{L} = \mathcal{L}_{\text{PDE}}$$

three competing terms under soft enforcement become one

- for $t > 0$ the spatial factor still vanishes on the walls
- each factor switches off exactly one condition
- the $(1 - t)$ factor assumes t in $[0, 1]$: scale the time axis first
- two loss terms disappear, and the slice is no longer outvoted

THE TRIAL SOLUTION, TERM BY TERM

$(1 - t) f_{\text{IC}}$ the initial field, fading with t

$t \cdot x(1-x) y(1-y)$ zero at $t = 0$ and on the walls

$\mathcal{N}(x, y, t)$ free everywhere else

soft three terms compete; the IC loses

hard one term; the IC is exact

When a condition lives on a lower-dimensional slice, penalising it is structurally weak. Build it in; Ex_07.2 notebooks 01 and 02 measure the difference.

Wave Equation

KIND, FUNCTION, CONDITIONS, APPLICATIONS

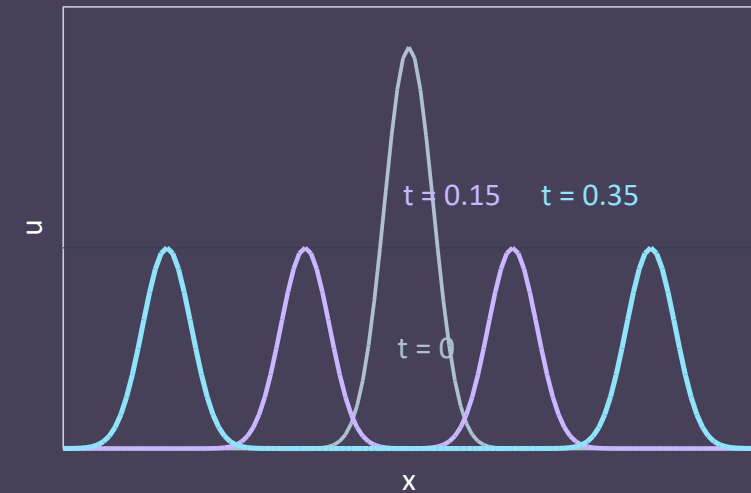
- **kind** — hyperbolic; second order in time
- **function** — propagation at a finite speed c
- **conditions** — the walls, the initial field, and the initial velocity
- **against heat** — features persist and travel instead of decaying
- **applications** — a struck panel, a string, acoustics, structural modes

$$u_{tt} = c^2 (u_{xx} + u_{yy}) + f(x, y, t)$$

second order in time, so two initial conditions: the displacement and the velocity

$$u(x, y, 0) = u_0(x, y), \quad u_t(x, y, 0) = v_0(x, y)$$

A PULSE SPLITS AND TRAVELS



d'Alembert's solution: the pulse splits into two halves that travel at c and keep their shape. Nothing decays, and that is what spectral bias finds hardest.

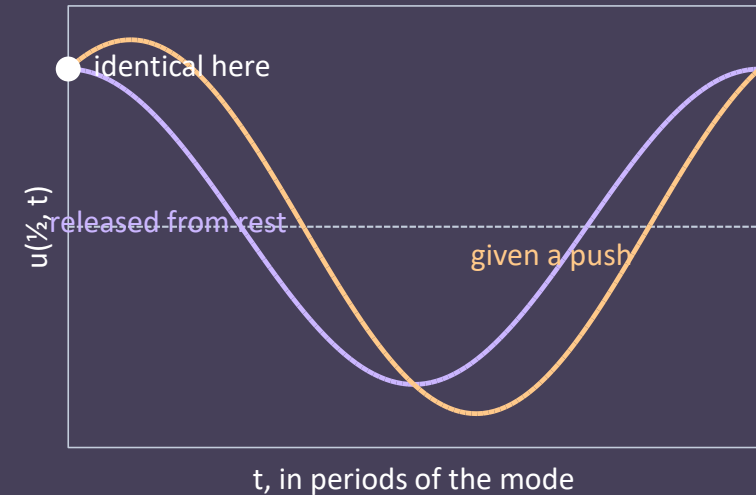
Two Initial Conditions

THE SAME SHAPE AT $T = 0$, TWO DIFFERENT FUTURES

- released from rest, or given a push: the same shape at $t = 0$
- displacement alone does not determine the motion
- without the velocity term the problem is under-determined
- $u \equiv 0$ satisfies the PDE, the walls and a flat initial field, exactly
- the residual converges, and the panel never moves

TALK · 2 MIN Ex_07.2 notebook 04 solves the panel correctly except for the velocity condition, and its loss converges better than the correct run. Which diagnostic catches it?

u AT THE CENTRE, AGAINST TIME



The mode $\sin \pi x$, either let go or pushed at $2 \sin \pi x$ per unit time. Same displacement at $t = 0$; the amplitude and the phase of everything after differ.

Periodic Activations

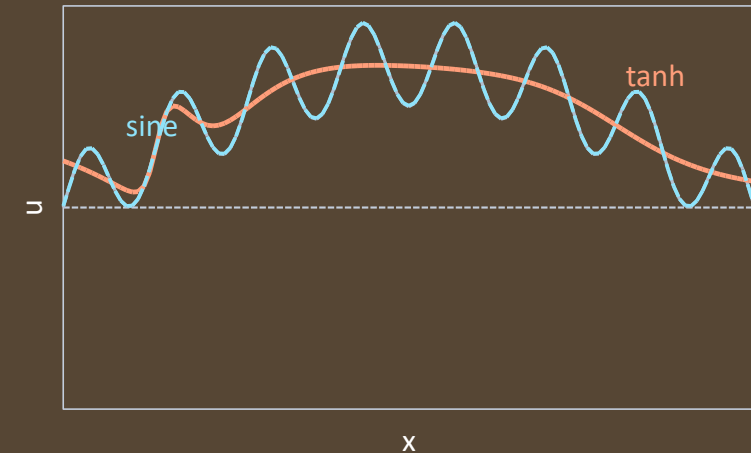
MATCH THE INDUCTIVE BIAS TO THE SOLUTION

$$\varphi(\mathbf{x}) = \sin(\omega_0 \mathbf{W}\mathbf{x} + \mathbf{b}), \quad \omega_0 \approx 30$$

SIREN: a sine first layer scaled by ω_0 , with a matched initialisation

- a wave solution is a sum of sinusoids: build the network from them
- the derivative of a sinusoid is a sinusoid, at every order
- the direct remedy for spectral bias on an oscillatory problem
- ω_0 sets the frequencies the first layer can reach at initialisation
- compare at equal parameter count, or you measure size

MEASURED · TANH AGAINST SINE



tanh 32 x 3

2209 weights, error 25.7 %

sine 32 x 3

2209 weights, error 0.53 %

Target $\sin(\pi x) + 0.3 \sin(15 \pi x)$, 3000 Adam steps, lr 1e-3, seed 0. Tanh recovers the envelope, not the detail.

Helmholtz Equation

KIND, FUNCTION, CONDITIONS, APPLICATIONS

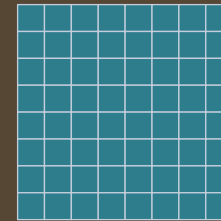
- **kind** — an eigenvalue problem: the mode shape and the scalar λ
- **function** — Poisson with the source replaced by λu
- **conditions** — the walls, a norm, orthogonality to lower modes
- **against Poisson** — u is on both sides; only certain pairs solve it
- **applications** — vibration and acoustic modes; separated variables

$$\nabla^2 u + \lambda u = 0, \quad \lambda = k^2$$

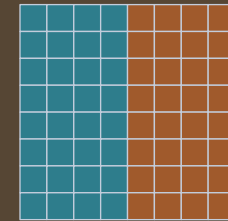
it arises whenever you separate variables in the wave or the heat equation

$$u_{mn} = 2\sin(m\pi x)\sin(n\pi y), \quad \lambda_{mn} = (m^2 + n^2)\pi^2$$

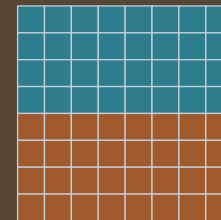
FIRST FOUR MODES OF THE UNIT SQUARE



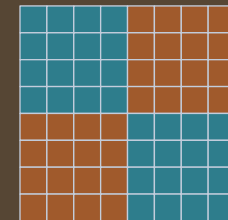
(1,1) $\lambda = 2\pi^2$



(2,1) $\lambda = 5\pi^2$



(1,2) $\lambda = 5\pi^2$



(2,2) $\lambda = 8\pi^2$

Two modes share $\lambda = 5\pi^2$: a degeneracy, and any combination of them is a mode too.

Trivial Solution

THE SOLUTION THE OPTIMISER FINDS FIRST

$$u \equiv 0, \quad \lambda = 0$$

satisfies the equation and the walls exactly, with a residual of zero

- gradient descent finds it reliably and quickly
- the expected outcome, not an occasional failure
- two safeguards together: a normalisation term and a floor on λ
- the first loss in this course that must exclude a solution

TALK · 2 MIN A zero residual is a perfect score for a wrong answer. For each of the four equations before this one, what else satisfies the loss you would write?

WHAT SATISFIES WHICH TERM

	residual	walls	norm = 1
$u \equiv 0, \lambda = 0$	yes	yes	no
mode (1,1)	yes	yes	yes
mode (2,1)	yes	yes	yes
$2 \times \text{mode (1,1)}$	yes	yes	no

λ trainable

one more parameter, beside $\hat{w}$

$\lambda \geq \lambda_0$

a floor, so zero is out of reach

norm = 1

so the scale is fixed

On the unit square the lowest eigenvalue is $2\pi^2 \approx 19.74$. Far from it is a bug.

Orthogonality and Normalisation

$$\langle u_{mn}, u_{pq} \rangle = \int_{\Omega} u_{mn} u_{pq} d\Omega = \delta_{mp} \delta_{nq}$$

distinct modes are orthogonal; without using it every run converges to the lowest

$$u \leftarrow u_{\text{raw}} - \sum_{k < K} \frac{\langle u_{\text{raw}}, u_k \rangle}{\langle u_k, u_k \rangle} u_k$$

projection — remove the lower modes from the ansatz: hard enforcement

$$\mathcal{L}_{\text{orth}} = \sum_{k < K} \langle u_{\hat{w}}, u_k \rangle^2$$

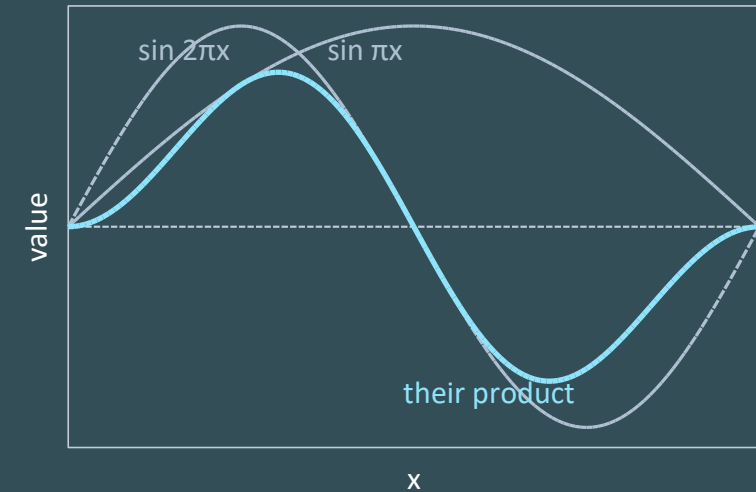
$$\mathcal{L}_{\text{norm}} = (\langle u_{\hat{w}}, u_{\hat{w}} \rangle - 1)^2$$

penalty — ask for orthogonality and for unit norm in the loss: soft enforcement

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + w_o \mathcal{L}_{\text{orth}} + w_n \mathcal{L}_{\text{norm}}, \quad \lambda \geq \lambda_0$$

- a mode is defined up to a scale: fix the norm, and λ is comparable
- one mode at a time, each run warm-started from the last

ORTHOGONALITY, VISUALLY



The product has equal positive and negative area, so its integral is zero. On an irregular domain that integral is the T3 quadrature again.

Burgers Equation

KIND, FUNCTION, CONDITIONS, APPLICATIONS

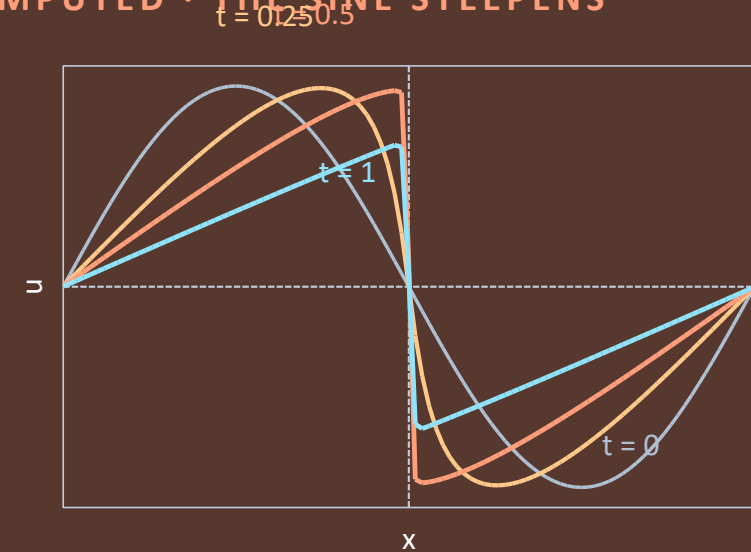
- **kind** — nonlinear; convection against diffusion, one dimension
- **function** — a velocity that carries itself: the term $u u_x$
- **conditions** — an initial sine, zero at both ends
- **against the wave** — transport steepens a smooth profile to a shock
- **applications** — traffic, gas dynamics, a prototype of Navier–Stokes

$$u_t + u u_x = \nu u_{xx}, \quad \nu = \frac{0.01}{\pi}$$

Raissi's setting: a viscosity of $0.01/\pi$, small enough that the front is nearly a jump

$$u(0, x) = -\sin(\pi x), \quad u(t, -1) = u(t, 1) = 0$$

COMPUTED · THE SINE STEEPENS



Integrated by a conservative finite-volume scheme, 800 cells. The slope at $x = 0$ grows from π to about 57 by $t = 0.5$, and the front is then a few cells wide.

Shock Formation

$$\frac{dx}{dt} = u$$

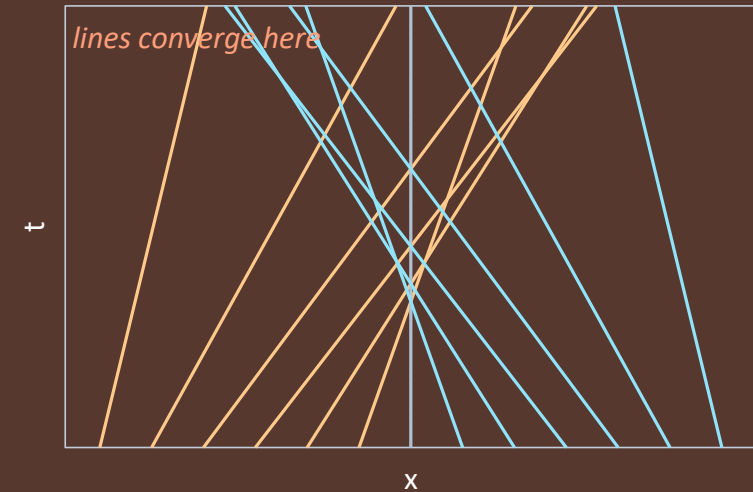
each value of u travels at its own speed along a characteristic, and where faster values catch slower ones the characteristics cross: a shock

WHY A SHOCK IS HARD FOR A PINN, AND WHAT TO DO

- **spectral bias, sharpened** — a near-jump is all high frequency
- **sampling imbalance** — the shock occupies almost no domain
- **large residual gradients** — u_{xx} is enormous at the front
- **a smooth ansatz** — tanh needs extreme weights to be steep
- **remedies, in order** — refine at the shock, more points, split time

TALK · 2 MIN Poisson needed no adaptive sampling and Burgers cannot do without it. In terms of where the residual is large, what changed?

CHARACTERISTICS IN THE x - t PLANE



Amber values move right, cyan values move left, and they meet at $x = 0$. A uniform sample spends almost every point where the solution is flat and already correct.

Discovering the Equation

THE INVERSE PROBLEM, TWO LINES OF CODE AWAY

$$f := u_t + \lambda_1 u u_x - \lambda_2 u_{xx}$$

λ_1 and λ_2 promoted to trainable parameters, beside the weights

$$\mathcal{L} = \frac{1}{N_u} \sum_{i=1}^{N_u} |u_{\hat{w}}(t_{u^i}, x_{u^i}) - u^i|^2 + \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_{f^i}, x_{f^i})|^2$$

a data term on N_u measurements of u , plus the residual on N_f collocation points

- the forward and inverse problems differ by two lines of code
- the data term is now essential: it is what identifies λ_1 and λ_2
- Raissi recovers $\lambda_1 = 1$ and $\lambda_2 = 0.01/\pi$ from scattered, noisy data
- the classical route is guess, solve, compare, repeat

REFERENCE CONFIGURATION

	Raissi et al. (2019)
data points N_u	2 000, scattered in x and t
collocation N_f	the same points
network	9 layers of 20, tanh
optimiser	L-BFGS, full batch
recovered λ_1, λ_2	within a fraction of a percent
with 1 % noise	still within a few percent

Same network, same loop. The gradient step that updates the weights updates the two coefficients as well, and that asymmetry, trivial here and expensive for a classical solver, is one of the strongest reasons to use the method.

Verification

$$\tilde{x} = \frac{x}{L}, \quad \tilde{t} = \frac{t}{\tau}, \quad \tilde{u} = \frac{u - u_0}{\Delta u}$$

non-dimensionalise first: lengths, times and the field to order one

FOUR CHECKS THE LOSS CANNOT MAKE

- **manufactured solution** — choose u , derive f , solve, compare
- **known eigenvalues** — $2\pi^2 \approx 19.74$ on the unit square, exactly
- **conservation** — insulated walls conserve energy; watch the drift
- **convergence with N** — refine the sample; a plateau is the network
- no error estimate of its own: no mesh to refine, no residual bound

THE DIE, BEFORE AND AFTER SCALING

	before	after
length	10 mm	1
time	$\tau = L^2/\alpha = 1.1 \text{ s}$	1
diffusivity	$8.8 \times 10^{-5} \text{ m}^2/\text{s}$	1
temperature	$300 \rightarrow 340 \text{ K}$	$0 \rightarrow 1$
the residual	kelvin per second	order one

A PINN that will not train often has nothing wrong with it except its units. The $(1 - t)$ trial solution assumed a unit time axis; the loss weights assumed comparable terms. Scaling gives both.

Challenges

THE RIGHT SHAPE, THE WRONG LEVEL

Poisson with Neumann walls only · the solution is determined up to a constant, and the residual is zero for all of them · pin one value, or add a mean constraint

LATE TIMES FIT, EARLY TIMES DO NOT

heat, soft initial condition · the slice is outvoted by the block · hard-enforce the initial field with the $(1 - t)$ factor, or weight early times

A SMOOTH ENVELOPE, NO DETAIL

wave, tanh activations · spectral bias on an oscillatory solution · periodic activations at equal size, and a denser sample

SYMPTOM AND FIX, BY EQUATION

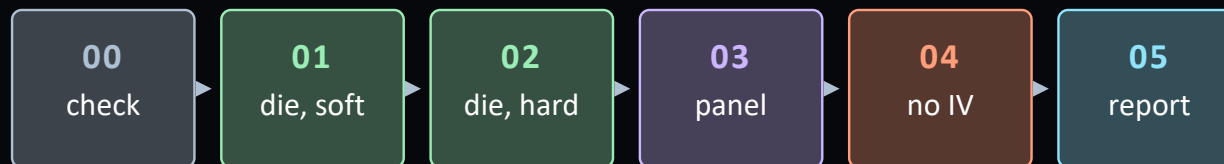
	symptom	fix
Poisson	right shape, wrong level	pin one value
heat	early times wrong	hard IC
wave	envelope only	sine activations
Helmholtz	$u \rightarrow 0$ and $\lambda \rightarrow 0$	norm term, floor on λ
Burgers	wiggles at the shock	refine, split time

Every symptom is visible in plots you already make: the per-term loss curves, and a residual map over the domain. Make both, every run.

Exercise 7.2

SIX NOTEBOOKS; A DIE AND A PANEL

- **00** — environment check: space–time sampling, derivatives in t
- **01** — the die, soft initial condition: where the error goes
- **02** — the die, hard initial condition: what the network pays
- **03** — the panel, both conditions, over three periods
- **04** — the panel that never moved: the velocity condition left out
- **05** — compare, and the report



THE TWO PROBLEMS

the die	10 × 10 mm silicon; two decaying modes
the panel	40 × 40 cm, 141 Hz, struck at 0.5 m/s
hand in	soft against hard IC; with and without IV
and	tanh against sine, at equal size

REPORT QUESTION

Notebook 04 converges better than notebook 03 and is wrong. Say why in terms of what the PDE requires, not of what the loss did.

Key Takeaways

- 1 **Classification — type dictates conditions**
The discriminant sorts a second-order equation into elliptic, parabolic or hyperbolic, and the type says which conditions make it well posed and therefore which loss terms you carry.
- 2 **Poisson — the reference**
A value or a flux at every boundary point, no gaps. On a rectangle a polynomial factor enforces the walls exactly; on a plate with a hole a small learned factor does the same.
- 3 **Heat — time is a coordinate**
One new term, one new input, one initial field on a slice that the block outvotes. The $(1 - t)$ trial solution makes the initial condition exact and removes two loss terms.
- 4 **Wave — two initial conditions**
Second order in time needs the displacement and the velocity. Leave the velocity out and $u \equiv 0$ satisfies everything else, and the loss converges. Sine activations match an oscillatory solution.
- 5 **Helmholtz — a zero residual is not enough**
The mode and the eigenvalue are unknown together, and $u \equiv 0$ with $\lambda = 0$ scores perfectly. Fix the norm, put a floor on λ , and project out the lower modes.
- 6 **Burgers — nonlinearity is free, the shock is not**
Autograd handles $u u_x$ without comment. What costs is the near-jump it creates: spectral bias, a sample that never sees it, huge residual gradients. Refine there. Two trainable coefficients discover the equation.
- 7 **Verification — scale first, then check**
Non-dimensionalise before anything else, then a manufactured solution, a known eigenvalue, a conservation law, convergence with N . A PINN gives no error estimate of its own.
- 8 **Challenges — a symptom per equation**
Wrong level, early times wrong, envelope only, collapse to zero, wiggles at the shock: each visible in the per-term loss curves and the residual map you make every run.

Next — L8.1: the heat equation in earnest, on real device geometry, with real units.

Questions

TEN QUESTIONS FOR THE ORAL EXAMINATION

- 1 What does the discriminant tell you about a second-order equation, and why does it matter for the loss? *Liu ch. 2, 4, 5*
- 2 Why does Poisson need no initial condition, and what goes wrong with Neumann conditions on every wall? *Liu ch. 2.4*
- 3 How is a Dirichlet condition enforced exactly on a domain with no polynomial factor? *Liu ch. 3.1 – 3.3*
- 4 What does a PINN do differently from a time-marching solver, and what does it give up? *Liu ch. 4.1 – 4.2; Wang et al. (2022)*
- 5 Why is the initial condition the term most often lost, and how does the (1 – t) trial solution fix it? *Liu ch. 4.2 – 4.3; Ex_07.2 notebooks 01 – 02*
- 6 Why does the wave equation need two initial conditions, and what does the panel do without the second? *Liu ch. 5.1 – 5.2; Ex_07.2 notebooks 03 – 04*
- 7 What does a sinusoidal activation buy on an oscillatory problem, and how must the comparison be made? *Liu ch. 5.7.4; Sitzmann et al. (2020)*
- 8 Why is $u \equiv 0$ a problem for Helmholtz and not for Poisson, and what two safeguards exclude it? *Liu ch. 6.2, 6.4.3*
- 9 What do orthogonality and normalisation add to the Helmholtz loss, and what is the hard form of each? *Liu ch. 6.4 – 6.5*
- 10 Why does Burgers need adaptive sampling when Poisson does not, and how does the inverse variant recover v ? *Raissi et al. (2019), §3.1, 4.1*

NEXT

L8.1 — Stationary Heat Transfer

The elliptic problem of today with real units, a convection boundary, a plate with a cooling hole, and an engineering question behind every solve: how hot does it get, and where?

BEFORE THE NEXT SESSION

read	Liu ch. 4.1 – 4.4, 5.1 – 5.4
run	Ex_07.2 notebooks 00 to 04
bring	the panel's loss curves, both runs
think about	which type your own problem is

Block L7 is finished: the method, and the five equations it is applied to. From here every lecture is one of these equations with a name and a unit.